

ReIG-DQ: An Audited Design Framework for Dissipative Quantum Dynamics

Compatibility Boundaries, Projective Convergence, Core Attraction, Logical
Recovery, and Ordered CPTP Paths

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Abstract

This paper presents the reconstructed ReIG-DQ design framework for dissipative quantum systems. The objective is not to claim a new universal physical principle, but to organize the finite-dimensional mathematical content that survived formal audit into distinct layers: compatibility, convergence, dissipative attraction, logical recovery, and finite-stage path dependence.

First, for a Lüders map associated with a finite-dimensional PVM, invariance of a single operator is fixed as equivalent to commutation with all measurement projectors. In contrast, operational compatibility of an entire core operator space is characterized not by pairwise commutation of every core projector with every measurement projector, but by invariance of the core operator space under the Lüders map. Bell-projector families provide explicit examples showing that the pairwise strong condition is sufficient but not necessary.

Second, the notion of identity for linear iteration is placed in complex projective space rather than in a phase-fixed vector representation. Under a strict dominant spectral gap, an algebraically simple dominant eigenvalue, and nonzero initial support in the dominant spectral subspace, normalized iterates converge to the dominant eigenray. For pure states,

$$D_{\text{tr}}(\rho_u, \rho_v) = \sin d_{\text{FS}}([u], [v]),$$

so ray convergence and rank-one density-operator convergence are equivalent representations of the same projective phenomenon.

Third, in a finite-dimensional GKLS model with a partial-isometry jump, we obtain an exact exponential attraction law for core occupancy I_0 , an exact block solution, an asymptotic CPTP idempotent retraction, and the fixed-point operator space. Core occupancy is kept explicitly distinct from logical identity I_1 .

Fourth, for a specified error isometry V , the recovery condition

$$K_R V = P$$

yields exact logical recovery on the full code operator algebra, a finite-time error/recovered mixture, an exact conditional logical identity channel, and

$$\frac{1}{2} \|\Lambda_t - \mathcal{J}_P\|_{\diamond} = e^{-\kappa t}.$$

Fifth, finite-stage ordered CPTP paths generated by two GKLS generators are formalized. Their generator commutator controls the leading short-time order sensitivity, but is not assumed to be a GKLS generator. In the canonical qubit witness, the commutator exponential is trace preserving but not completely positive. Moreover, exact reachability in a same-primitive baseline is tautological and is not treated as evidence of irreducibility, non-simulability, or N2 novelty.

The central contribution is therefore not theorem proliferation. It is an auditable design discipline that separates **what has been proved from what has not been proved**, using identity levels, physical-map constraints, provenance, and an explicit Quarantine Register.

Keywords: dissipative quantum dynamics, GKLS, CPTP, projective convergence, quantum error recovery, Lüders map, path dependence, provenance, audit

1. Introduction

Frameworks that attempt to integrate quantum systems, open-system dynamics, and control semantics are vulnerable to a recurring class of category errors: treating occupancy of a subspace as preservation of logical information; treating convergence of a vector as convergence of its physically phase-insensitive state; or treating order dependence between finite-time operations as evidence for a new physical principle.

ReIG-DQ was reconstructed to prevent those conflations. Rather than defending a broad set of earlier claims, the reconstruction retains only the mathematics that survived audit and recasts it as finite-dimensional theorem packs with explicit scope and failure boundaries.

The present paper is organized around five frozen mathematical cores:

1. **T-DQ-01** — operational compatibility boundary;
2. **T-DQ-02** — projective asymptotic convergence;
3. **T-DQ-03** — I_0 core attraction;
4. **T-DQ-04** — I_1 exact logical recovery;
5. **T-DQ-05** — finite-stage path semantics.

An Identity Bridge separates core membership I_0 , logical identity I_1 , projective identity I_{ray} , and the rank-one density representation of a pure state.

This paper does **not** claim a new universal quantum principle. The novelty level is N1. The constituent mathematics belongs to known finite-dimensional linear algebra, projective geometry, GKLS dynamics, quantum channels, quantum error correction, and noncommuting finite-time control. The contribution of ReIG-DQ is the audited integration of these structures together with explicit identity tags, scope boundaries, and a reproducible provenance chain.

2. Reconstruction Method and Provenance

2.1 Quarantine-first reconstruction

The reconstruction procedure prioritizes quarantine over rhetorical salvage. A historical claim that is false, unsupported, physically inadmissible, or stronger than the available evidence is not silently weakened and reintroduced under a new label.

In particular, the following failure modes are not reused:

- treating non-CPTP maps as physical open-system dynamics;
- identifying F_{core} with logical fidelity;
- using a conserved observable as a transition order parameter;
- upgrading noncommutativity by itself to N2 novelty;
- using a baseline that already contains the target construction as evidence of non-reducibility;
- inferring a universal coherent/classical dichotomy from a local example.

2.2 Builder / Independent Verifier / Freeze

Each mathematical core was fixed by the following workflow:

1. a Builder formulates a minimal-scope theorem candidate;
2. an Independent Verifier attempts to break it using counterexamples, numerical implementations, and boundary cases;
3. apparent contradictions are analyzed to distinguish theorem defects from verifier or numerical-instrument defects;
4. theorem or verification code is corrected as needed;
5. canonical artifacts and SHA-256 hashes are frozen.

Agreement between multiple AI systems is not treated as mathematical proof. The relevant evidence is the ordinary mathematics together with a provenance record showing which claim was checked by which artifact, which verifier-side error occurred, and which corrected artifact became canonical.

2.3 Authority order

If the manuscript and supplementary artifacts appear to disagree, authority is resolved in the following order:

1. frozen theorem pack;
2. freeze manifest;
3. canonical verifier report / corrected verifier script;
4. integration / publication / canonical ledger;
5. historical / pre-fix artifact.

This order prevents a known-bug pre-fix verifier from later being treated as canonical evidence.

3. Identity Semantics

3.1 Four semantic levels

The word “identity” is not used without a level tag.

I_0 : core membership

For a projector P ,

$$\text{Tr}(P\rho) = 1$$

means that the state is supported in the designated core subspace. This is an occupancy statement.

I_1 : logical identity

The encoded logical content is preserved. This is stronger than core membership.

I_{ray} : projective identity

For a pure-state vector v , the physical state modulo global phase is

$$[v] \in \mathbb{P}(\mathcal{H}).$$

Rank-one density representation

For a pure normalized vector,

$$\rho_v = |v\rangle\langle v|$$

is a one-to-one density-operator representation of the same projective pure state.

3.2 Frozen non-implications

The Identity Bridge fixes the guardrails

$$\boxed{I_0 \not\Rightarrow I_1}$$

and

$$\boxed{I_0 \not\Rightarrow I_{\text{ray}}}.$$

A core can contain multiple logical states and multiple pure-state rays. Therefore core membership alone does not determine internal logical or projective identity.

3.3 Ray/density correspondence for pure states

For normalized pure states u, v ,

$$D_{\text{tr}}(\rho_u, \rho_v) = \sqrt{1 - |\langle u, v \rangle|^2},$$

while

$$d_{\text{FS}}([u], [v]) = \arccos |\langle u, v \rangle|.$$

Hence

$$\boxed{D_{\text{tr}}(\rho_u, \rho_v) = \sin d_{\text{FS}}([u], [v])}.$$

Thus ray convergence and rank-one density convergence are equivalent for pure states. A mixed state, in general, does not admit a unique single-ray representation.

4. T-DQ-01: Operational Compatibility Boundary

4.1 Lüders fixed points

Let $\{Q_\alpha\}$ be a PVM on a finite-dimensional Hilbert space, with

$$Q_\alpha Q_\beta = \delta_{\alpha\beta} Q_\alpha, \quad \sum_\alpha Q_\alpha = I.$$

Following the historical context of the Lüders projective measurement update [1], define

$$\mathcal{E}_Q(X) = \sum_\alpha Q_\alpha X Q_\alpha.$$

For any $X \in \mathcal{B}(\mathcal{H})$, T-DQ-01A states the equivalence of

$$\mathcal{E}_Q(X) = X,$$

$$Q_\alpha X Q_\beta = 0 \quad (\alpha \neq \beta),$$

$$X = \sum_\alpha Q_\alpha X Q_\alpha,$$

and

$$[X, Q_\alpha] = 0 \quad (\forall \alpha).$$

Therefore, for a single projector Π ,

$$\boxed{\mathcal{E}_Q(\Pi) = \Pi \iff [\Pi, Q_\alpha] = 0 \quad (\forall \alpha)}.$$

4.2 Core extraction and operational compatibility

For mutually orthogonal core projectors $\{\Pi_k\}$, define

$$\mathcal{P}_{\text{core}}(X) = \sum_k \Pi_k X \Pi_k$$

and

$$\mathcal{A}_{\text{core}} = \text{Ran } \mathcal{P}_{\text{core}} = \bigoplus_k \Pi_k \mathcal{B}(\mathcal{H}) \Pi_k.$$

On Hilbert-Schmidt operator space, $\mathcal{P}_{\text{core}}$ and $\mathcal{E}_Q$ are orthogonal projections. T-DQ-01B gives the equivalence

$$\boxed{\mathcal{P}_{\text{core}} \mathcal{E}_Q = \mathcal{E}_Q \mathcal{P}_{\text{core}}}$$

and

$$\boxed{\mathcal{E}_Q(\mathcal{A}_{\text{core}}) \subseteq \mathcal{A}_{\text{core}}}.$$

The relevant boundary is therefore closure/invariance of the core operator space, not completeness of the core projector family.

4.3 The pairwise strong condition is sufficient, not necessary

If

$$[\Pi_k, Q_\alpha] = 0 \quad (\forall k, \alpha),$$

then operational compatibility follows. However, this condition is not necessary.

For the complete Bell PVM

$$\{\Pi_{\Phi^+}, \Pi_{\Phi^-}, \Pi_{\Psi^+}, \Pi_{\Psi^-}\}$$

against a local computational PVM, each Bell projector violates the pairwise strong condition, with

$$\max_{\alpha} \|[\Pi_k, Q_\alpha]\|_{2 \rightarrow 2} = \frac{1}{2}.$$

Nevertheless,

$$\mathcal{E}_Q(\Pi_{\Phi^\pm}) = \frac{1}{2}(\Pi_{\Phi^+} + \Pi_{\Phi^-}),$$

and

$$\mathcal{E}_Q(\Pi_{\Psi^\pm}) = \frac{1}{2}(\Pi_{\Psi^+} + \Pi_{\Psi^-}),$$

so the Bell-diagonal core algebra is invariant and

$$\boxed{\mathcal{P}_{\text{Bell}} \mathcal{E}_Q = \mathcal{E}_Q \mathcal{P}_{\text{Bell}}}.$$

Thus pairwise strong compatibility is sufficient but not necessary.

This result is not attributed to Lüders [1]; [1] is used only for historical context for the projective measurement update.

4.4 Operational incompatibility boundary

Define

$$\delta_{\text{inv}} = \|(I - \mathcal{P}_{\text{core}})\mathcal{E}_Q\mathcal{P}_{\text{core}}\|_{2 \rightarrow 2}.$$

If $\delta_{\text{inv}} > 0$, the core operator space is not invariant and

$$\mathcal{P}_{\text{core}}\mathcal{E}_Q \neq \mathcal{E}_Q\mathcal{P}_{\text{core}}.$$

This is an operational incompatibility statement. It is not a universal POVM or instrument no-go theorem, nor does it imply destruction of every possible notion of identity.

5. T-DQ-02: Projective Asymptotic Convergence

5.1 Dominant eigenray theorem

Let M be a linear map on a finite-dimensional complex Hilbert space. Assume that it has an algebraically simple eigenvalue $\lambda_1 \neq 0$ satisfying the strict dominance condition

$$|\lambda_1| > r := \max_{\lambda \in \sigma(M) \setminus \{\lambda_1\}} |\lambda|,$$

and let P_1 be the Riesz spectral projection of λ_1 . Assume

$$P_1 v_0 \neq 0.$$

For the normalized iteration

$$v_n = \frac{M^n v_0}{\|M^n v_0\|},$$

one has

$$\boxed{[v_n] \longrightarrow [u_1]}$$

or equivalently

$$\boxed{d_{\text{FS}}([v_n], [u_1]) \rightarrow 0.}$$

Here u_1 is a normalized vector spanning $\text{Ran } P_1$.

Write

$$v_0 = c u_1 + w, \quad c \neq 0.$$

Then

$$M^n v_0 = c \lambda_1^n u_1 + N^n w,$$

where N is the restriction to the non-dominant spectral complement. Because the spectral

radius of N is $r < |\lambda_1|$,

$$\frac{\|N^n w\|}{|\lambda_1|^n} \rightarrow 0.$$

Since multiplication by a nonzero scalar does not change a ray, projective convergence follows.

5.2 Jordan-corrected rate

Let m be the largest Jordan-block size among non-dominant spectral blocks on $|\lambda| = r$. Then

$$\|N^n\| = O(n^{m-1} r^n),$$

and

$$d_{\text{FS}}([v_n], [u_1]) = O\left(n^{m-1} \left(\frac{r}{|\lambda_1|}\right)^n\right).$$

If the non-dominant restriction is diagonalizable, $m = 1$, and the rate reduces to a purely geometric envelope.

5.3 Vector convergence versus ray convergence

Write

$$\lambda_1 = |\lambda_1| e^{i\theta}.$$

After normalization, one can have

$$v_n = e^{i(n\theta + \phi_0)} u_1 + o(1).$$

Therefore the ray can converge while the raw vector does not. After phase alignment,

$$e^{-i(n\theta + \phi_0)} v_n \rightarrow u_1.$$

The self-convergence object used in ReIG-DQ is therefore the projective ray rather than a phase-fixed vector.

This projective treatment is consistent with the use of projective Hilbert-space geometry and the Fubini–Study metric in Anandan and Aharonov [2]. T-DQ-02A itself, however, is formulated independently as the frozen finite-dimensional spectral theorem above and is not directly derived from [2].

5.4 No-gap boundary

The absence of a strict gap does not imply that every ray fails to converge. With distinct equal-modulus eigenvalues, a generic initial state can fail to converge because of persistent relative phase, while an initial eigenray remains fixed.

The correct boundary statement is

absence of a strict gap removes the universal convergence guarantee.

5.5 T-DQ-02E

DEFERRED / SUPPORTED EXAMPLE ONLY

For a defective dominant Jordan block, numerical and analytic examples support an $O(1/n)$ -type projective convergence. A general defective-dominant theorem is **not** part of the frozen T-DQ-02 main claim and is not promoted in this paper.

6. T-DQ-03: I_0 Core Attraction

6.1 One-way partial-isometry GKLS model

Let P be the core projector and $Q = I - P$. Let the jump operator satisfy

$$K = PKQ, \quad K^\dagger K = Q.$$

Define

$$\mathcal{L}_K(\rho) = \kappa \left(K\rho K^\dagger - \frac{1}{2}\{Q, \rho\} \right).$$

This is a standard GKLS generator. Its placement within completely positive quantum dynamical semigroup theory is consistent with the foundational results of Lindblad [3] and Gorini-Kossakowski-Sudarshan [4].

6.2 Exact I_0 attraction

Define the occupancy

$$F_{\text{occ}}(\rho) = \text{Tr}(P\rho).$$

T-DQ-03A gives

$$\mathcal{L}_K^\dagger(P) = \kappa Q,$$

hence

$$\frac{d}{dt}F_{\text{occ}} = \kappa(1 - F_{\text{occ}}).$$

Therefore

$$F_{\text{occ}}(t) = 1 - (1 - F_{\text{occ}}(0)) e^{-\kappa t}$$

and

$$F_{\text{occ}}(t) \rightarrow 1.$$

The identity tag of this theorem is

$$\boxed{I_0 \text{ only}}.$$

6.3 Exact block solution

With respect to the P/Q block decomposition,

$$\boxed{\begin{aligned} \Phi_t(\rho_0) = & P\rho_0P + e^{-\kappa t/2}(P\rho_0Q + Q\rho_0P) \\ & + e^{-\kappa t}Q\rho_0Q + (1 - e^{-\kappa t})K\rho_0K^\dagger. \end{aligned}}$$

Thus

$$\Phi_t(\rho_0) \rightarrow \mathcal{R}_K(\rho_0) = P\rho_0P + K\rho_0K^\dagger.$$

The asymptotic map has Kraus operators $\{P, K\}$, is CPTP, and satisfies

$$\boxed{\mathcal{R}_K^2 = \mathcal{R}_K}.$$

6.4 Fixed-point space

For the minimal generator,

$$\boxed{\text{Fix}(\mathcal{L}_K) = P\mathcal{B}(\mathcal{H})P}.$$

Hence every state supported in the core is pointwise fixed in this minimal model. This is not generalized to arbitrary core-attracting GKLS generators.

6.5 General I_0 Lyapunov sufficient condition

For a finite-dimensional GKLS generator $\mathcal{L}$, if

$$\mathcal{L}^\dagger(P) \succeq \gamma Q, \quad \gamma > 0,$$

then

$$\boxed{1 - F_{\text{occ}}(t) \leq e^{-\gamma t}(1 - F_{\text{occ}}(0)).}$$

This is a sufficient all-state I_0 -attraction criterion. An expectation inequality verified only on a restricted state set is insufficient for a uniform all-state guarantee, as shown by the trap-state counterexample in the frozen verification package. We do **not** claim that the operator-order condition is the unique weakest possible assumption among all alternative formulations.

Historical work on engineered dissipation, including quantum reservoir engineering by Poyatos, Cirac, and Zoller [5] and dissipation-driven state engineering by Verstraete, Wolf, and Cirac [6], provides context for using dissipation as a design resource. The exact ReIG-DQ I_0 law, however, is not attributed to those works.

7. T-DQ-04: I_1 Exact Logical Recovery

7.1 Code/error isometry

Let P be the code projector and let V satisfy

$$V^\dagger V = P, \quad VV^\dagger = Q, \quad PQ = 0.$$

Define

$$K_R = V^\dagger.$$

Then

$$\boxed{K_R V = P.}$$

If $V = EP$ for a physical error E , this becomes

$$\boxed{K_R EP = P.}$$

7.2 Exact I_1 recovery

For every code-supported logical operator

$$X = PXP,$$

one has

$$\boxed{K_R V X V^\dagger K_R^\dagger = X.}$$

In particular, for every encoded density operator $\rho_L = P\rho_L P$,

$$\boxed{K_R V \rho_L V^\dagger K_R^\dagger = \rho_L.}$$

The identity tag is

$$\boxed{I_1}.$$

7.3 Dissipative recovery mixture

Define

$$\mathcal{L}_R(\rho) = \kappa \left(K_R \rho K_R^\dagger - \frac{1}{2} \{Q, \rho\} \right).$$

For the corrupted logical state

$$\rho_{\text{err}} = V \rho_L V^\dagger,$$

the exact finite-time state is

$$\Phi_t(\rho_{\text{err}}) = e^{-\kappa t} V \rho_L V^\dagger + (1 - e^{-\kappa t}) \rho_L.$$

Thus the logical state is recovered asymptotically.

7.4 Conditional versus unconditional quantities

The code branch is

$$P \Phi_t(V \rho_L V^\dagger) P = p_t \rho_L, \quad p_t = 1 - e^{-\kappa t}.$$

Therefore the success probability is input independent:

$$p_{\text{succ}}(t) = 1 - e^{-\kappa t}.$$

For $t > 0$, the normalized successful branch is exactly

$$\Lambda_{t,\text{cond}} = \text{Id}_{\text{logical}}$$

and hence

$$F_{e,\text{cond}} = 1.$$

For the unnormalized successful branch,

$$F_{e,\text{uncond}} = 1 - e^{-\kappa t}.$$

Conditional exact recovery is not the same as deterministic finite-time recovery.

7.5 Residual diamond distance

The physical output channel is

$$\Lambda_t(\rho_L) = (1 - q_t) \rho_L + q_t V \rho_L V^\dagger, \quad q_t = e^{-\kappa t}.$$

The completed-recovery target is $\mathcal{J}_P(\rho_L) = \rho_L$. Since the code and error outputs occupy orthogonal sectors,

$$\frac{1}{2} \|\Lambda_t - \mathcal{J}_P\|_\diamond = e^{-\kappa t}$$

and, in the full-norm convention,

$$\|\Lambda_t - \mathcal{J}_P\|_\diamond = 2e^{-\kappa t}.$$

7.6 Canonical two-qubit example

Let

$$P = |00\rangle\langle 00| + |11\rangle\langle 11|, \quad E = I \otimes X.$$

Then

$$V = EP = |01\rangle\langle 00| + |10\rangle\langle 11|,$$

and

$$K_R = V^\dagger = |00\rangle\langle 01| + |11\rangle\langle 10|.$$

Hence

$$K_R EP = P.$$

For arbitrary

$$|\psi_L\rangle = \alpha|00\rangle + \beta|11\rangle,$$

one has

$$K_R E |\psi_L\rangle = |\psi_L\rangle.$$

This preserves the complete logical superposition, not merely code occupancy.

For general exact quantum error correction, Knill and Laflamme [7] provide the standard necessary-and-sufficient perfect-recovery framework. T-DQ-04 does **not** replace or extend the general Knill-Laflamme conditions; it is restricted to one specified error isometry.

8. T-DQ-05: Finite-Stage Path Semantics

8.1 Ordered physical paths

For two finite-dimensional GKLS generators $\mathcal{L}_C, \mathcal{L}_I$ and stage times $t_C, t_I \geq 0$, define

$$\Phi_{CI} = e^{t_I \mathcal{L}_I} e^{t_C \mathcal{L}_C}$$

and

$$\Phi_{IC} = e^{t_C \mathcal{L}_C} e^{t_I \mathcal{L}_I}.$$

Each semigroup and each ordered composition is CPTP.

8.2 Global order-independence criterion

The following are equivalent:

$$[\mathcal{L}_I, \mathcal{L}_C] = 0$$

and, for every $t_C, t_I \geq 0$,

$$\Phi_{CI}(t_C, t_I) = \Phi_{IC}(t_C, t_I).$$

Equality at one isolated time pair is not used as a converse proof of generator commutation.

8.3 Leading order sensitivity

As $t_C, t_I \rightarrow 0$,

$$\Phi_{CI} - \Phi_{IC} = t_C t_I [\mathcal{L}_I, \mathcal{L}_C] + O((|t_C| + |t_I|)^3).$$

The commutator is a local diagnostic of order sensitivity. It is not treated as a physical GKLS generator by default.

8.4 Commutator-not-GKLS guardrail

For the canonical qubit raising/lowering example, let

$$\mathcal{C} = [\mathcal{L}_I, \mathcal{L}_C].$$

Then

$$e^{0.1\mathcal{C}}$$

is trace preserving but not completely positive. In the canonical verifier implementation, the minimum Choi eigenvalue is approximately

$$-0.0350.$$

This number is a verifier witness, not a new theorem. The relevant guardrail is

$$\text{trace preservation} \not\Rightarrow \text{complete positivity.}$$

Therefore an independent GKLS check is required before a generator commutator could be interpreted as a physical semigroup generator.

8.5 State-observable witness

Let

$$\Delta_{CI} = \Phi_{CI} - \Phi_{IC}.$$

At fixed stage times, the following are equivalent:

1. $\Delta_{CI} \neq 0$;
2. there exists a density operator ρ such that $\Delta_{CI}(\rho) \neq 0$;
3. there exist a density operator ρ and Hermitian observable O such that

$$\text{Tr}[O\Phi_{CI}(\rho)] \neq \text{Tr}[O\Phi_{IC}(\rho)].$$

A channel-level order-sensitivity metric can be defined by

$$\delta_\diamond = \frac{1}{2} \|\Phi_{CI} - \Phi_{IC}\|_\diamond.$$

The canonical verifier's Choi half-norm value 0.1266 is only a supporting finite-time witness

and is **not** called a diamond norm.

8.6 Exact reachability with the same primitives

Let

$$\mathfrak{M}(\mathcal{L}_C, \mathcal{L}_I)$$

be the control monoid generated by finite compositions of the two primitive semigroups. By definition,

$$\boxed{\Phi_{CI}, \Phi_{IC} \in \mathfrak{M}(\mathcal{L}_C, \mathcal{L}_I).}$$

Therefore a baseline that permits the same primitives and arbitrary finite-stage ordering reaches the target path with exact approximation error

$$\boxed{0}.$$

This is tautological reachability. It is **not** evidence of N2 novelty, irreducibility, or non-simulability.

8.7 Identity-level path diagnostics

Path ordering can change

$$\Delta I_0(\rho) = \text{Tr}[P\Phi_{CI}(\rho)] - \text{Tr}[P\Phi_{IC}(\rho)].$$

However, ΔI_0 does **not** imply any I_1 conclusion.

An I_1 path diagnostic requires an explicit code, encoding, error/recovery model or logical readout, and logical target channel.

I_{ray} is used only if both compared outputs are pure. If either output is mixed, a density-operator metric must be used instead.

Thus

$$\boxed{\text{order sensitivity} \neq \text{new physical principle}.}$$

Related Works

The references in this section are used to position the frozen ReIG-DQ results within the existing literature. They are not used to transfer external authority to ReIG-DQ, and the frozen theorems are not described as being derived from these references unless that is literally true.

Measurement update and compatibility

Lüders [1] provides the historical source for the projective measurement state-update rule. The T-DQ-01 core-operator-space invariance criterion, Bell-family counterexample, and the non-necessity of the pairwise strong condition are independently formalized in the frozen ReIG-DQ theorem pack and are **not attributed to [1]**.

Projective geometry of pure quantum states

Anandan and Aharonov [2] provide important background for projective Hilbert-space geometry and the Fubini-Study metric. The phase-insensitive ray representation used in T-DQ-02 is consistent with that geometric setting. The strict-dominant-gap theorem, the $P_1 v_0 \neq 0$ condition, and the Jordan-corrected rate are independently stated in the frozen T-DQ-02 pack and are **not presented as directly derived from [2]**.

GKLS dynamics and engineered dissipation

The standard completely positive quantum dynamical semigroup framework is situated within the foundational work of Lindblad [3] and Gorini-Kossakowski-Sudarshan [4]. T-DQ-03 through T-DQ-05 remain inside that standard GKLS setting.

For engineered dissipation, Poyatos, Cirac, and Zoller [5] and Verstraete, Wolf, and Cirac [6] provide historical and engineering context for treating dissipation as a design resource. The exact ReIG-DQ I_0 law, the I_0/I_1 guardrail, and the T-DQ-05 order-sensitivity boundary are **not attributed to those references**.

Quantum error correction

Knill and Laflamme [7] provide the standard general framework of necessary and sufficient conditions for perfect quantum error recovery. T-DQ-04 treats one specified error isometry and **is not a replacement for, or extension of, the general Knill-Laflamme conditions**.

Positioning rule

The references provide context, not borrowed authority. Mathematical authority remains with the frozen theorem packs and the Canonical Ledger.

9. Quarantine Register and Recurrence Prevention

The reconstruction does not merely delete failed historical claims; it converts them into explicit recurrence-prevention rules.

9.1 F_{core} versus identity

Core occupancy is I_0 ; logical identity is I_1 .

$$I_0 \not\Rightarrow I_1.$$

Exact core attraction can succeed while an additional logical unitary corrupts the encoded content. This prevents occupancy fidelity from being silently reinterpreted as logical fidelity.

9.2 Non-CPTP maps

Physical dissipative dynamics are represented by GKLS/CPTP maps. T-DQ-05 retains the explicit witness that a commutator exponential can be TP but non-CP, preventing the inference “trace preserving implies physical.”

9.3 Noncommutativity and novelty

Neither

$$[\mathcal{L}_I, \mathcal{L}_C] \neq 0$$

nor

$$\Phi_{CI} \neq \Phi_{IC}$$

establishes more than order sensitivity. They do **not** by themselves imply a new cooperation variable, universal law, or irreducibility.

9.4 Self-containing baseline

If the primitive baseline already contains the target path as one of its allowed compositions, exact approximation error is zero by definition. This is not novelty evidence.

9.5 Conserved observable as order parameter

The previously quarantined SWAP-expectation construction is not revived as a transition order parameter.

10. Discussion

10.1 Why the five cores remain distinct

The five frozen mathematical cores are complementary rather than interchangeable.

T-DQ-01 concerns compatibility between measurement/coarse-graining structure and a core operator space.

T-DQ-02 concerns projective identity and asymptotic convergence of normalized linear iteration.

T-DQ-03 concerns dissipative transfer of population into a core subspace, i.e. an I_0 phenomenon.

T-DQ-04 concerns preservation and recovery of encoded logical content, i.e. an I_1 phenomenon.

T-DQ-05 concerns finite-time order sensitivity under differently ordered CPTP stages.

This separation prevents multiple physical and mathematical meanings from being compressed into a single scalar fidelity or an unlabeled use of the word “identity.”

10.2 Role of the audit history

The provenance record preserves not only final passing results but also intermediate misclassifications and verifier defects. Examples include:

- overextending pairwise compatibility from sufficient to necessary;
- misidentifying completeness as the compatibility boundary;
- misclassifying a dominant-Jordan example because of an insufficient time horizon;
- a numerical spectral-projection artifact for a non-normal matrix;
- conflation of I_0 and I_1 ;
- insufficiency of a restricted-set expectation Lyapunov condition;
- a factor-of-two/channel-dimension implementation error in diamond-distance verification;

- recognizing that a trace-preservation test alone does not establish complete positivity of a commutator exponential.

These episodes are not cited to claim that “multiple AIs agreed, therefore the theory is correct.” They are preserved so that a third party can trace where independent verification paths encountered the same defect boundaries, and how each issue was resolved mathematically or numerically.

10.3 Position as a design framework

ReIG-DQ does not propose a new fundamental physical law. It supplies a structural design discipline for keeping distinct evaluation layers separate:

compatibility → identity semantics → core attraction
→ logical recovery → ordered path semantics.

A pass at one layer is not automatically promoted into a pass at the next layer.

11. Limitations

The results of this paper have explicit scope.

1. Finite-dimensional setting

Infinite-dimensional generalizations are outside the present work.

2. T-DQ-01 uses PVMs and Lüders maps

General POVMs and non-projective instruments require separate analysis.

3. T-DQ-02 assumes an algebraically simple dominant eigenvalue

The general defective-dominant theorem is not frozen.

T-DQ-02E = DEFERRED / SUPPORTED EXAMPLE ONLY.

4. Pointwise-fixed core in T-DQ-03 is a property of the minimal model

It is not claimed for arbitrary core-attracting GKLS generators.

5. T-DQ-04 treats one specified error isometry

It does not replace general quantum error-correction conditions for arbitrary error sets.

6. T-DQ-05 concerns order sensitivity

Noncommuting paths are not upgraded to a new cooperation principle, non-simulability result, or N2 novelty claim.

7. Mixed states and rays

No extension assigning a unique projective ray to an arbitrary mixed state is used.

8. Novelty level

The paper remains N1. No new universal physical principle is claimed.

12. Conclusion

The reconstruction of ReIG-DQ prioritizes mathematically valid finite-dimensional cores and their boundaries over preservation of broader historical claims.

T-DQ-01 characterizes operational compatibility by invariance of a core operator space and shows that pairwise commutation is not necessary.

T-DQ-02 replaces phase-fixed vector identity with projective ray identity and gives its exact pure-state density representation.

The Identity Bridge separates I_0 , I_1 , and I_{ray} .

T-DQ-03 formalizes I_0 core attraction as exact GKLS dynamics.

T-DQ-04 adds the recovery condition $K_R V = P$ and gives a separate I_1 exact logical recovery theorem.

T-DQ-05 formalizes finite-stage ordered CPTP dynamics and freezes the guardrail

order sensitivity $\neq$ new physical principle.

The central result of the reconstruction is therefore not the number of new claims. It is the creation of a traceable mathematical structure in which compatibility, identity, attraction, recovery, and path dependence each have separate acceptance conditions.

ReIG-DQ is positioned as an auditable dissipative-quantum design framework that preserves both what can be claimed and what must not yet be claimed.

Appendix A. Frozen theorem mapping

Manuscript section	Frozen authority
§3 Identity semantics	Identity Bridge v1.0 + T-DQ-02C
§4 Operational compatibility	T-DQ-01 v1.0
§5 Projective convergence	T-DQ-02 v1.0
§6 Core attraction	T-DQ-03 v1.0
§7 Logical recovery	T-DQ-04 v1.0
§8 Finite-stage paths	T-DQ-05 v1.0
§9 Quarantine	Canonical Ledger v1.0
§10–12 framing/limits	Publication Architecture v1.0 + Canonical Ledger v1.0

Appendix B. English synchronization audit requirements

The independent English-version audit should verify at least:

1. every frozen boxed equation matches the Japanese bibliography-frozen manuscript;
2. all T-DQ-02A hypotheses are preserved;
3. T-DQ-02E remains DEFERRED / SUPPORTED EXAMPLE ONLY;
4. $I_0 \neq I_1$ and $I_0 \neq I_{\text{ray}}$ remain explicit;
5. conditional and unconditional quantities remain distinct;
6. both diamond-norm conventions remain explicit;
7. 0.1266 remains explicitly non-diamond;
8. all seven DOI strings match the Japanese References exactly;
9. attribution disclaimers remain strong: not attributed to, not directly derived from, not a replacement for;
10. no N2/new-universal-principle assertion enters through translation;
11. SWAP appears only in quarantine context;
12. the phrase order sensitivity $\neq$ new physical principle remains semantically explicit.

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Synchronization provenance

This English v0.1 candidate is synchronized from the Japanese bibliography-frozen manuscript. It intentionally preserves: - frozen mathematics; - identity-level distinctions; - bibliography numbering and DOI strings; - attribution disclaimers; - T-DQ-02E deferred status; - the diamond-norm factor-of-two convention; - the 0.1266 non-diamond label; - the N1 novelty discipline.

It remains **unfrozen** until an independent Japanese-English synchronization audit passes.

Synchronization Freeze Provenance

The independent Japanese-English synchronization audit passed all seven required checks:

1. frozen boxed equations are present in both languages;
2. key numerical values remain synchronized;
3. all seven DOI strings are identical;
4. no borrowed-authority or ungated N2 wording entered through translation;
5. attribution disclaimers remain explicit in English;
6. T-DQ-02E deferred status, the 0.1266 non-diamond label, and identity tags remain intact;
7. the three T-DQ-02A hypotheses remain intact.

The initial `en_sync_audit.py` produced two apparent failures caused by verifier-side pattern limitations: a literal 0.5000 search and a narrow negation window for English not X, Y, or Z constructions. `en_check4_redo.py` and the final audit report resolved those artifacts. No manuscript correction was required.

This v1.0 artifact is therefore synchronized and frozen against the Japanese bibliography-frozen manuscript.